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A preliminary paleomagnetic pole for mid-Cretaceous rocks from Tobago: further evidence for large clockwise rotations in the Caribbean–South American plate boundary zone

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Abstract

The island of Tobago occupies the eastern end of the central (igneous) belt of the Caribbean–South American plate boundary zone. Volcaniclastic sediment of the Albian (~ 100 Ma) Tobago Volcanic Group and dikes of similar age within it were sampled in two homoclinal sections with different attitudes. The mean of virtual geomagnetic poles for 12 sites (25.4°N, 24.1°E, $A_{95} = 4.2°$) is well defined, pre-tilting and apparently reliable, yet is far removed from a similar age reference pole for South America. Five other paleomagnetic studies of Cretaceous rocks from widely separated localities farther west in the plate boundary zone yield pole positions that are remarkably similar to the Tobago pole. Poles obtained from the Guajira Peninsula of Colombia, the islands of Aruba and Bonaire, and the Caribbean Mountains of Venezuela are among those that agree with the Tobago result. The paleolatitudes for study areas within the plate boundary zone are consistent with an origin on or near South America, yet the poles throughout the zone are rotated roughly 90°. Dextral relative motion between the Caribbean and South American plates was probably responsible for the rotation.

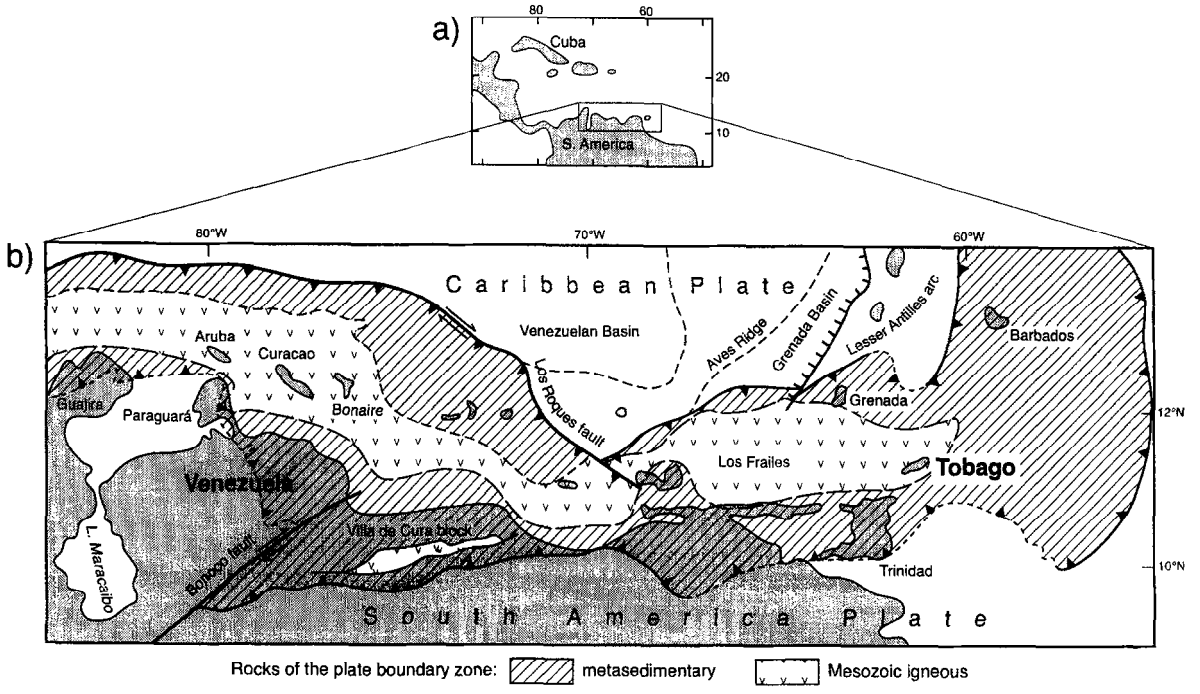
Keywords: Tobago; Albian; paleomagnetism; pole positions; Tobago Volcanic Group; Tobago Intrusive Group

1. Introduction

Paleomagnetic studies detect latitudinal displacements and vertical-axis rotations, hence they can make valuable contributions to the analysis of plate kinematics and plate margin tectonic processes. The small paleomagnetic data set that exists for localities within the Caribbean–South America plate boundary zone has been interpreted as recording large amounts of clockwise rotation [1–3]. Here we report results from a reconnaissance study of mid-Cretaceous rocks from Tobago, West Indies, which suggest that large clockwise rotations are common to the entire length of the Caribbean–South American plate boundary zone.

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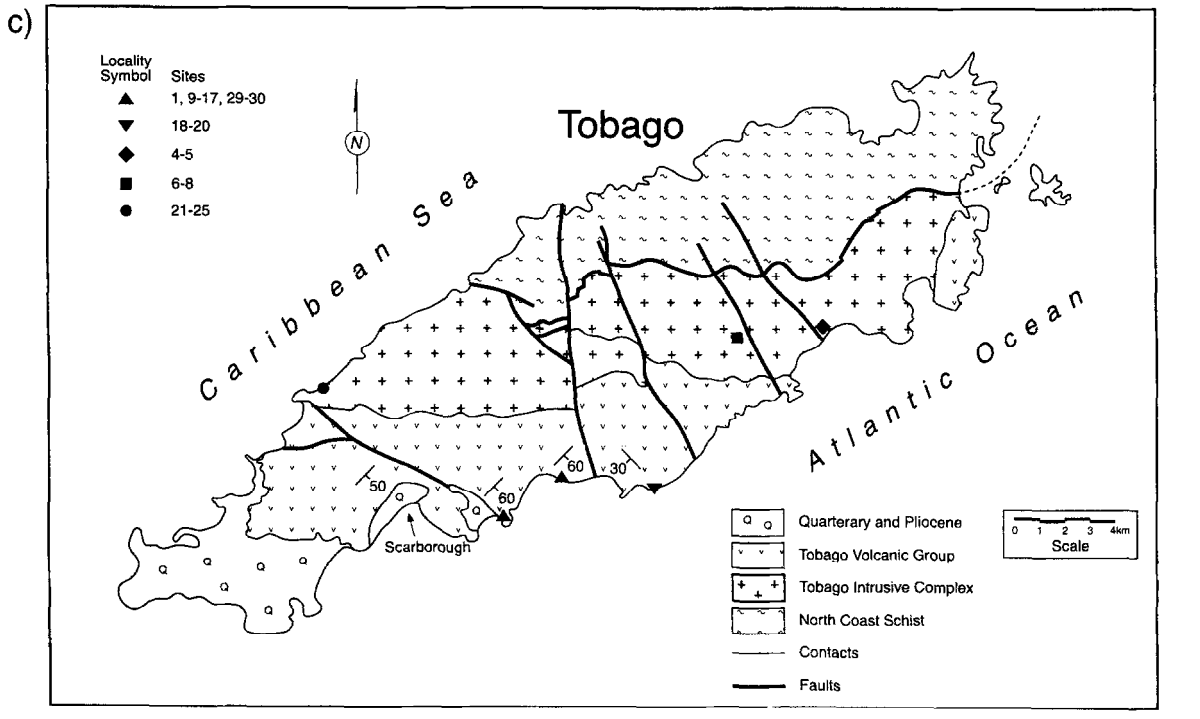


Fig. 1. (a) Location map. (b) Caribbean–South American Plate boundary zone and locations of other paleomagnetic studies. (c) Geology of Tobago (modified from [10]) and paleomagnetic localities reported on here. Symbols for separate localities are also used in the other figures. Only sites 29 and 30 come from the southwest of the two localities marked by upright triangles.

2. Region setting and local geology

The Caribbean plate comprises oceanic lithosphere of the Venezuelan and Colombian Basins; its age and place of origin are uncertain. It may be a far-travelled piece of the Farallon or some other Pacific basin plate (e.g., [4]), or it may be of more local origin. Between the Caribbean and South American plates is a broad plate boundary zone, about 250 km wide, that includes a central belt of igneous rocks of Cretaceous age (pillow basalt, island-arc volcanic and plutonic rocks; Fig. 1b), flanked by belts of deformed sedimentary rocks. The Villa de Cura block, consisting of island-arc metavolcanic rocks and intrusions, is apparently an outlier of the central belt located within the southern flanking sedimentary belt. The central (igneous) belt comprises terranes that are exotic to South America [5,6]. Occupying the east end of this belt is the Tobago terrane [7].

The island of Tobago exposes Mesozoic igneous rocks in two distinct units ([6,8]; Fig. 1c). One is the Tobago Volcanic Group (TVG), consisting of pillow lavas, breccia, and volcanogenic turbidites. One lithic clast, low in this unit, yielded a 108 ± 3 Ma $^{40}Ar/^{39}Ar$ plateau age on hornblende [9], which is

Table 1
Paleomagnetic results from the Tobago Volcanic Group

Site	Du	Iu	Dt	It	N/No	R	α_{95}	k	BDAZ	Min	Max	Ty	Fit	NRM	Lithology
					Southeast-dipping section - Triangle										
01	40.2	+14.6	54.4	+12.2	6/6	5.978	4.5	227	135/55	20	90	mT[1]	f	0.061	mudstone
09	34.4	+19.8	60.4	+14.7	5/6	4.988	4.3	320	134/72	20	80	mT	f	0.004	siltstone
10	35.7	+18.5	57.5	+15.2	4/6	3.997	3.0	910	134/64	20	60	mT	a	0.005	mudstone
11	24.3	+29.8	68.2	+26.4	5/5	4.855	14.8	28	134/70	20	60	mT	a	0.50	dike
12	33.8	+19.9	60.4	+15.3	6/6	5.957	6.2	116	134/72	20	70	mT	a	0.008	mudstone
13	66.1	+28.0	66.1	+28.0	6/7	5.912	9.0	57	129/71	20	75	mT	f	0.16	greywacke
14	35.7	+40.0	77.1	+21.5	5/6	4.976	6.3	165	132/63	400	535	°C	f	0.14	mudstone
15[2]	11.3	+27.3	56.8	+35.5	6/6	5.763	14.9	21	130/65	20	80	mT	f	0.27	volc. breccia
16	38.5	+31.0	68.1	+13.8	4/5	3.965	9.9	86	130/65	20	70	mT	a	0.11	mudstone
17	46.3	+35.3	74.7	+9.4	4/4	3.987	6.1	225	131/67	500	570	°C	a	0.10	mudstone
29[2]	25.1	+37.2	58.4	+17.5	5/6	2.286		1	115/60	20	80	mT	a	0.60	volc. breccia
30	40.0	+37.2	64.2	+7.1	5/7	4.996	2.6	891	115/60	505	555	°C	a	0.75	dike
					Southwest-dipping section - Inverted Triangle										
18[3]	88.8	-15.4	88.4	+10.9	4/5	3.811	23.8	16	248/28	260	500	°C	a	0.001	mudstone
19	71.6	-3.3	72.0	+24.6	4/5	3.958	11.0	71	248/28	430	550	°C	a	0.011	mudstone
20	66.7	-0.2	66.5	+27.8	6/6	3.868	11.0	38	248/28	500	550	°C	a	1.36	dike
Mean	40.9	23.7			12/15	11.33	11.0	16							
			65.7	18.1		11.84	5.3	68							

Du, Iu, Dt, It = declination and inclination of site-mean remanence directions before (u) and after (t) correction for tilt. N/No = number of samples used in calculations/number collected (samples were discarded if broken, incoherently magnetized, or gave suspiciously discordant directions). R = resultant of N unit vector sample directions; α_{95} and k = radius of circle of 95% confidence and precision parameter [22], respectively. BDAZ = the azimuth and plunge of bedding down-dip direction. Min and Max = demagnetization range; Ty = type of demagnetization, mT = alternating field in milliTesla, °C = thermal. a = anchored fit; f = free least-squares line fit used to derive sample directions [18]. NRM = initial moment per unit volume (in A/m). [1] a.f. demagnetization followed thermal demagnetization to 360°C. [2] Not used because blocks sampled appear to retain remanence acquired before deposition. [3] Not used because of weak magnetization and poorly defined direction.

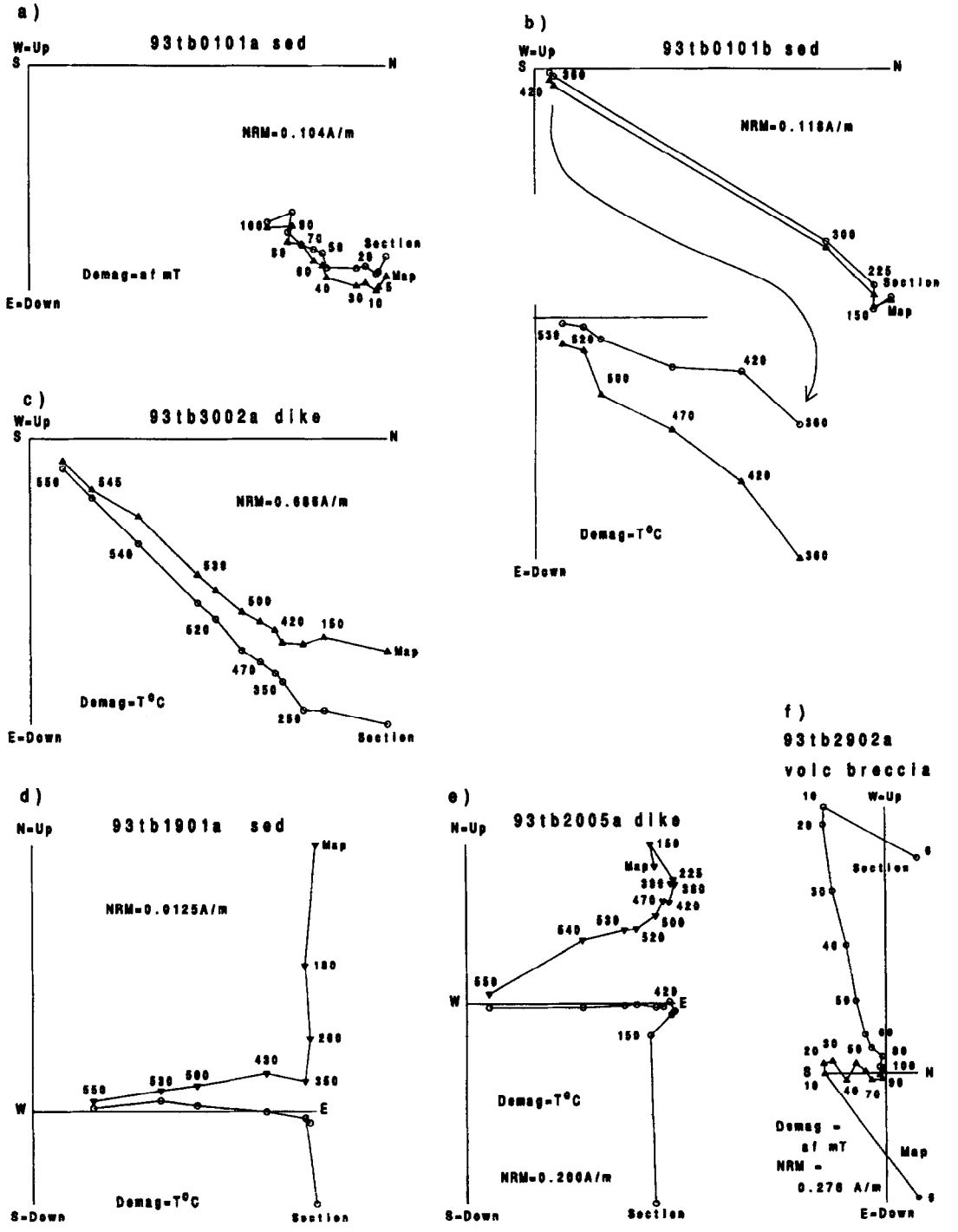


Fig. 2. Orthogonal diagrams showing demagnetization paths for sedimentary and dike rock from southeast-dipping block (a–c,f) and southwest-dipping block (d,e). High coercivity (a) but low unblocking temperature (b) suggest pyrrhotite carries much of the remanence in some sedimentary rocks but is absent from the dikes (c,e). Magnetizations with odd directions and coercivity (f) and unblocking temperature spectra appropriate for magnetite characterize volcanic bombs at site 29. ▲ = projections on map and ○ = projections on vertical section of vector end points after demagnetization steps indicated by labels.

consistent with the Albian age indicated by ammonite molds and radiolaria in an epiclastic interval. The other unit is the Tobago intrusive complex (TIC), with $^{40}Ar/^{39}Ar$ plateau ages on hornblende of coarse-grained phases ranging from 107 ± 3 to 102 ± 3 Ma, and on mafic dikes from 103 ± 3 to 91 ± 2 Ma [9]. Fission tracks in zircons indicate rocks in the TIC cooled below 250°C at about 103 Ma [10]. We sampled the TVG away from intrusive contacts where it is at prehnite–pumpellyite grade or lower [11]. Dikes from the TIC that invade the TVG were also sampled. In addition, we sampled a few sites within the TIC. The bedded TVG is exposed in homoclines with gentle to steep dips, separated by late Neogene normal faults. These homoclines may simply be tilted fault blocks. Structure within the TIC can only be inferred from regional considerations, but is likely to include tilting of fault blocks.

3. Previous paleomagnetic results

Paleomagnetic data from the central belt of the plate boundary zone imply large clockwise rotations of Aruba, Bonaire and at least some parts of the Villa de Cura body [1,2], and possibly Curacao and Los Frailes [12] as well. A similar amount of rotation was also found in the Mesozoic La Teta lava of the Guajira peninsula [13]. The Santa Ana intrusive complex of the central belt of Paraguaná gave highly scattered directions, with an approximately 180° declination anomaly [1]. Still other paleomagnetic studies [14] have been performed in the southern Caribbean region. On Tobago, paleomagnetic directions were previously determined from the TIC [12,15] and the Hawk's Bill Formation in the western part of the island [15,16]. The dacitic flows and pyroclastic rocks of the Hawk's Bill Formation may be equivalent in age to the youngest rocks of the TVG but have few reliable bedding indicators. Uncorrected directions from these studies are extremely scattered. They will be reviewed in the discussion, along with results for rocks of Mesozoic age that satisfy a set of minimal reliability criteria: means must have $\alpha_{95} < 20°$ and be derived from a minimum of 3 determinations of independent directions after a.f. or thermal cleaning [3].

4. A paleomagnetic reconnaissance of Tobago

4.1. Tobago Volcanic Group

We report here on samples from 15 sites (4–7 samples per site) in sediments of the TVG unit (Fig. 1c) and dikes within it. Results are summarized in Table 1. Reliable bedding indicators were difficult to recognize and were mainly found in a single area; thus our sampling sites are less widely spaced that we would have liked. Cores were oriented using both sun and magnetic compass; no significant local magnetic anomalies were encountered. Preparation, measurement, alternating field and thermal demagnetization, extraction of directions using least-squares line fitting over 4–10 points, and calculation of mean directions followed standard procedures (e.g., [17]). A few samples with poor line fits (MAD > 10°; [18])

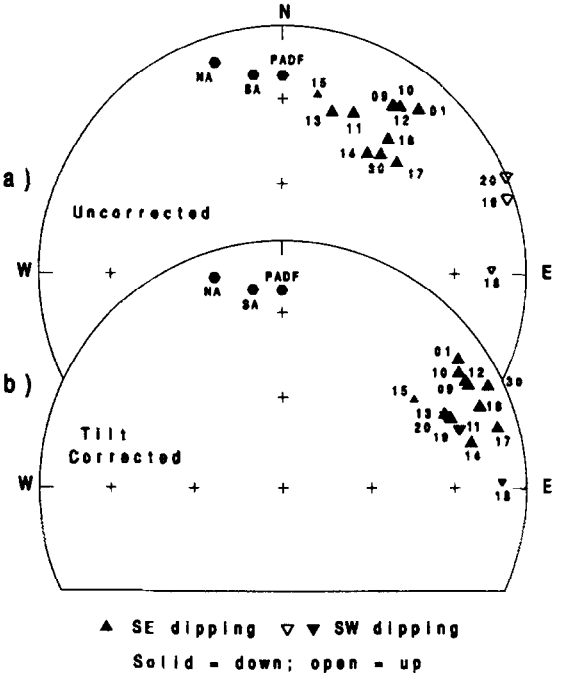


Fig. 3. Equal area projection of accepted site-mean directions from Table 1: (a) uncorrected and (b) tilt-corrected. Sites omitted from calculation of mean are shown with small symbols. Hexagons show directions expected for present axial dipole field (PADF) and mid-Cretaceous fields for North (NA; [19]), rotated into South American coordinates using [20], and South America (SA; from [19]).

were rejected, as was one site (#18) for which $\alpha_{95} > 20°$. Two sites (#15 and #29) obtained from volcanic breccias were also not used in later analysis. Samples from separate volcanic bombs at site #29 had magnetizations with stability characteristics similar to samples from accepted sites but widely scattered directions (Fig. 2f). This is most consistent with preservation of remanence acquired by the bombs before their deposition. Magnetizations of finer grained, tuffaceous site #15 are not as scattered, but are suspected of retaining some predeposition remanence. Six other sites were sampled but not fully processed because they showed little promise during pilot demagnetization.

Samples were collected in two homoclinal sections (triangles and inverted triangle, Fig. 1c) that had different dips: southeast on the west side of a north–south trending fault and southwest on the east side. Stepwise a.f. and thermal demagnetization indicate the following:

1. A northeast and down component dominates the magnetization of most sites from the southeast-dipping section (Fig. 2a–c).
2. Multicomponent but readily resolvable magnetiza-

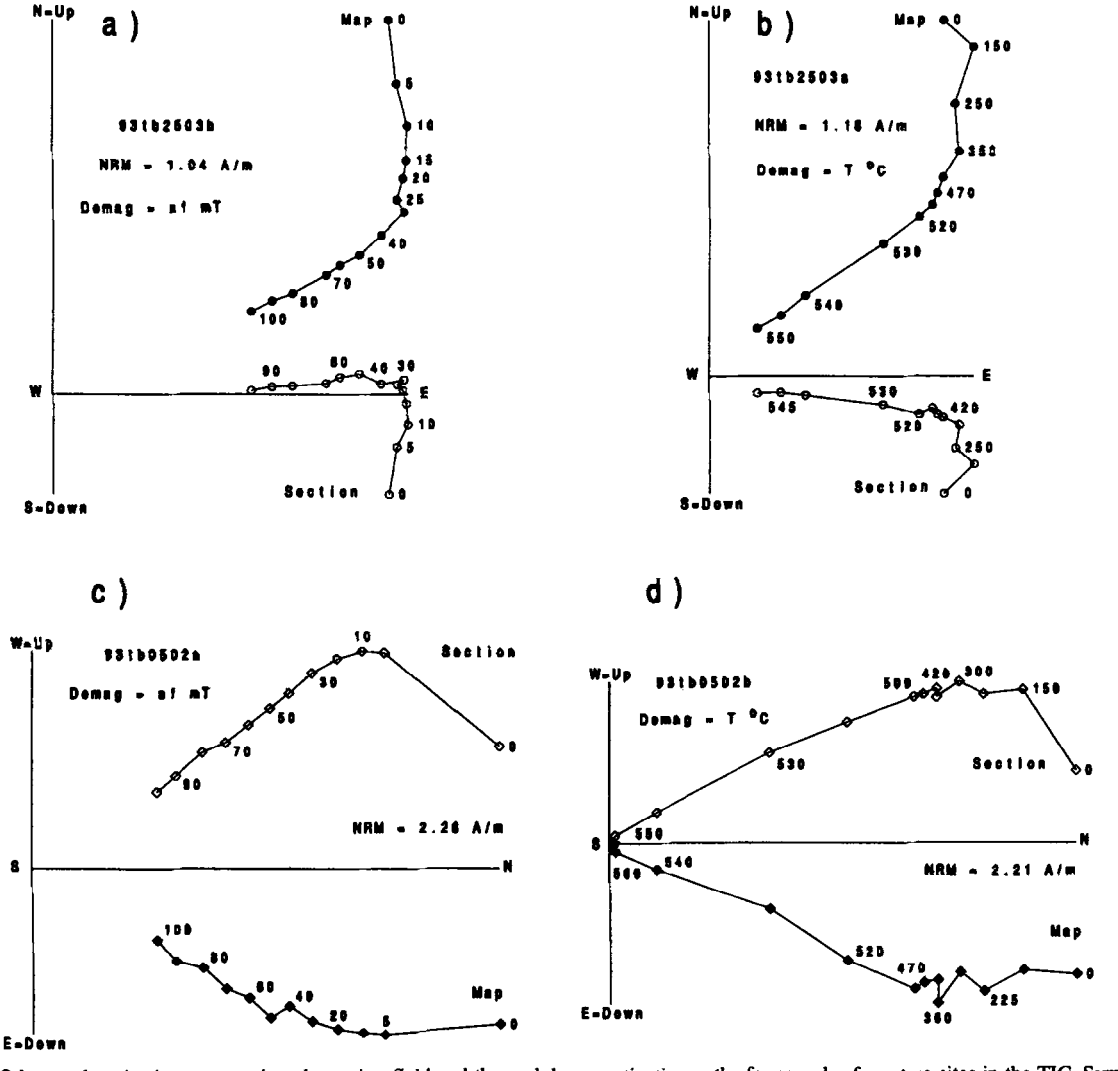


Fig. 4. Othogonal projections comparing alternating field and thermal demagnetization paths for samples from two sites in the TIC. Symbols as in Fig. 2.

tion is found throughout the southwest-dipping section, with the highest stability component being easterly and shallow (Fig. 2d,e).

3. Most fine-grained sedimentary rocks have a high coercivity, moderate unblocking temperature component that probably resides in pyrrhotite (Fig. 2a,b).
4. Demagnetization characteristics of dikes are consistent with magnetite being the only magnetic mineral (Fig. 2c,e).
5. A north-directed, gently downward, lower stability component is present in both dikes and sedimentary rocks of the southwest-dipping section (Fig. 2d,e).
6. A high-stability (characteristic) component found in most samples has an easterly declination that is quite different from reference directions calculated using reference poles for either cratonal North [19] or South America [19,20] (Fig. 3a).

Pyrrhotite is present, but seems to be confined to the sedimentary rocks. It is probably authigenic, and in the southeast-dipping section appears to retain the same direction, as does magnetite (Fig. 2b). Nevertheless, we worried that some pyrrhotite might represent late hydrothermal alteration; consequently, we derived sample directions only after thermal demagnetization above 350°C from sedimentary rocks that appeared to contain pyrrhotite. Site-mean directions

Table 2
Uncorrected site mean directions from the Tobago Intrusive Complex and Hawk's Bill Formation

Site	Du	Iu	N/No	R	α_{95}	k	Min	Max	Ty	Fit	NRM	lithology
Easternmost locality - Diamond Symbol												
04	16.7	-23.4	6/6	5.975	4.7	200.8	30	100	mT	f	0.51	diorite
05	21.6	-34.9	5/7	4.954	8.3	86.3	30	100	mT	f	1.50	diorite
East-central locality - Square Symbol												
06	56.4	-10.8	7/8	6.909	7.5	65.9	450	555	°C	a	0.22	diorite
07	51.6	-12.7	6/7	5.812	13.2	26.5	450	550	°C	a	0.18	diorite
08	60.2	-4.8	4/4	3.933	13.8	45.0	450	565	°C	f	1.62	dike
Westernmost locality - Round Symbol												
21	2.2	+19.6	6/6	5.946	7.0	92.6	530	625	°C	a	0.19	diorite
22	29.5	-17.1	8/7	7.961	4.1	180.5	30	90	mT	a	0.09	tonalite
23	29.1	-10.2	7/7	6.971	4.2	204.6	30	100	mT	a	0.07	dike
24	2.9	+15.1	5/6	4.959	7.8	96.6	20	80	mT	a	0.39	dike
25	51.3	+2.9	5/6	4.830	16.1	23.5	35	100	mT	a	1.04	dike
Previous results from TIC in the east - Right pointing triangle												
34	27	+32	5/5		11	53		50	mT		0.8	diorite
35	12	-8	7/7		23	8		30	mT		0.4	diorite
36	313	+42	4/4		17	30		50	mT		1.0	diorite
Y24	18	-17	2/2	2.0				20	mT			diorite
Hawk's Bill Formation in the west near 21-25 - Left pointing triangle												
T1	335.2	+23.2	3	2.959	17.9	48.6		30	mT		1	dacite?
T4	295.2	+2.1	3	2.933	23.0	29.8		30	mT		1	dacite?
T7	175.4	+10.3	3	2.993	7.1	300.8		30	mT		1	dacite?
T8	336.3	-11.0	3	2.981	12.2	103.7		30	mT		1	dacite?
Y22	350	-9	4	3.90				80	mT			dacite

Headings as in Table 1. 04–25, this study; 34–36 are from [12]; Y24, Y22 are sites Y29024 and Y29022 from [15]; T1–T8 are from [16].

used in the following analysis should thus reflect only magnetization carried by magnetite.

To investigate the age of the high stability remanence direction we performed a fold test [21]. Site-mean directions before tilt-correction form two distinct groups (Fig. 3a). Correction produces dramatic convergence, and improvement in the statistic k [22] that is significant at the 99% confidence level [23] (Fig. 3b; Table 1). In the analysis that follows we use the paleomagnetic pole calculated from site-mean directions after this simple tilt correction.

The fold test described above shows that the characteristic magnetization was acquired before tilting, but of course does not prove that the bedding was horizontal at the time of magnetization [24]. Application of reversal and baked contact tests to check for remagnetization is hampered by the age of the rocks, which is within the mid-Cretaceous interval of normal polarity, and by the short time between sedimentation and essentially coeval igneous activity. However, the stable but scattered directions from site #29 can be interpreted as a positive conglomerate test [21], which indicates that at least that site was not remagnetized after deposition. The fission track age of 103 Ma on zircons from the TIC [10] shows that wholesale thermal remagnetization after that time is unlikely. Furthermore, the low grade of metamorphism and lack of cleavage strongly suggest that the rocks have always existed at shallow depth and low temperature. The weight of this evidence suggests the rocks sampled were not significantly tilted before magnetization, which was likely original or nearly so. Thus, the mean direction after correction for tilt should provide a valid estimate of the geomagnetic field direction in Tobago during the Albian. Note that magnetization during this time of normal polarity removes any ambiguity about the sense of discordance of Tobago's pole from reference poles.

4.2. Tobago Intrusive Complex and published results

Largely to improve geographic coverage, we sampled at 10 sites from 3 localities within the Tobago

Table 3
Seleced Cretaceous poles from the Caribbean–South American plate boundary zone, reference poles for South America, and rotation and poleward motion implied by their discordance

Pole	Location	Plat°N	Plon°E	A_{95}	Age	R	P	Rel
A1	Aruba	68.6	28.4	4.1°	<77	22±5	0±5	ET
A2	"	4.8	16.9	?	77	97	4	LK
A3	"	18.2	28.8	18.5°	77	81±20	12±20	LK
B1	Bonaire	-6.3	32.7	15.0°	65	94±16	20±15	ET
C1	Guajira	9.9	20.8	14.5°	140	81±15	18±15	EK
T	Tobago	25.4	24.1	4.2°	105	74±6	0±6	MK
V1	N. Venezuela	58.8	113.8	17.0°	<100	8±19	27±18	LK
V2	"	24.7	102.2	?	<100	23	60	LK
V4	"	-4.2	14.7	12.4°	100	104±13	2±13	MK
ET	N. America	86.3	120.0	3.2°	60			
LK	Patagonia	78.7	178.4	6.3°	72			
MK	N. America	81.5	204.0	3.5°	102			
EK	Parana basin	83.8	274.6	3.4°	128			

Plat, Plon = latitude and longitude of pole; A_{95} = semi-angle of 95% confidence about pole. Age is estimated age of magnetization in Ma. R, P, Rel = rotation and poleward motion [31] relative to reference pole. A1 = Aruba batholith, 'predominant' direction (unblocked above 525°C) [2]. A2 = Aruba batholith, 'soft' direction (unblocked below 400°C) [2]. A3 = Aruba felsic dikes cutting batholith [2]. B1 = Washikemba Fm. [2]. C1 = La Teta lavas [13]; majority of sites interpreted here as reversly magnetized. T, = Tobago Volcanic Group, from corrected mean direction, Table 2. V1 = Paraguaná gabbroic and volcanic rocks [1]. V2 = Basic intrusions, Falcon Basin [1]. V4 = Villa de Cura block, including El Chaco ultramafic intrusion [1]. Reference poles: ET, 59–67 Ma pole [30] rotated to South America using Anomaly 25 reconstruction (59 Ma) from [20]. LK, Patagonia basalts [26]. LK, 88–124 Ma pole [19] rotated to South America using 84 Ma total reconstruction and half of 84–119 Ma stage reconstruction [20]. EK, Serra Geral basalts [27]. Some entries recalcualted by [3].

Intrusive Complex. Sample localities in dikes and massive diorite/tonalite are shown by round, square and diamond symbols in Fig. 1c. Samples responded well to magnetic cleaning (Fig. 4; Table 2), although most showed multiple components. The similarity in behavior between TIC site #25 (Fig. 4a,b) and TVG sites #19 and #20 (Fig. 2d,e) suggests that the two localities had similar magnetic histories, despite their geographic separation, lithologic differences, and location on separate fault blocks. Means of well-defined characteristic directions from all but two of the TIC sites have northeast declinations and are strongly divergent from present day or Cretaceous expected directions (Fig. 5a). Some have shallow inclinations found within the TVG (compare Figs. 5 and 3a). We are encouraged to find that the easterly directions from the TVG are not merely a local phenomenon, although we are unable to use the TIC results without structural control.

Previous results from Tobago show large scatter. Two sites from [12] and [15] show directions similar to sites from our eastern TIC localities and one has a direction similar to the TVG from the southeast-dipping section (Table 2; Fig. 5b). Quite different directions were obtained from another eastern site and all sites in the Hawk's Bill Formation from near our western TIC locality (Table 2; Fig. 5b; [15,16]). These, with one exception, show upward inclinations or counterclockwise declinations. The exception (T7) may be a reverse polarity remanence [16]. If so, it is possible that the magnetization in the Hawk's Bill Formation is of different (younger) age than the magnetization in the other units. The overall scatter of uncorrected directions from Tobago probably reflects severe structural complexity. Similarly scattered directions from plutons and country rock in the Omineca Belt of British Columbia were used to quantify tilt of extensional fault blocks [25]. Calculation of tilt in that study was facilitated by geological evidence that allowed the contributions that poleward displacement and vertical axis rotation made to the observed paleomagnetic discordance to be discounted. Since we can make no such simplifying assumptions here, we must abandon the uncorrectable directions and base the following analysis solely on results from the layered TVG sections. However, we interpret the scatter of uncorrected directions as indicating stability of magnetization from all rocks sampled at least since Neogene(?) tilting.

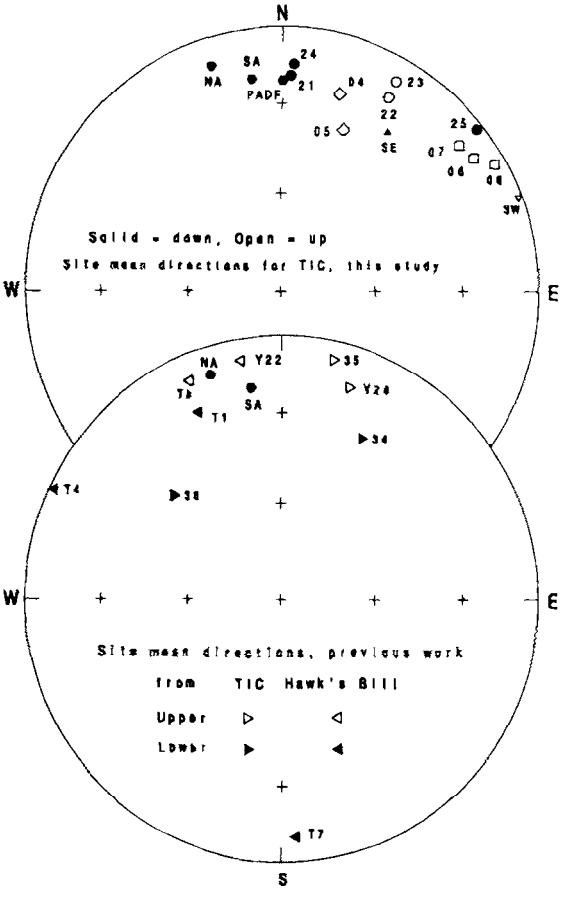


Fig. 5. Equal area projection of site-mean directions for the TIC and Hawk's Bill Formation from Table 2. Symbols as in Fig. 1 and Fig. 4. For reference, uncorrected locality mean directions for the SW- and SE-dipping sections of TVG are shown with small triangles. Reference directions (hexagons) as in Fig. 3.

5. Discussion

The paleomagnetic test of whether two crustal blocks have moved with respect to one another consists of comparing coeval paleomagnetic poles. To test whether a block has moved with respect to an adjacent stable continental craton one normally compares a pole from the block with a coeval point on the path of apparent polar wander (APW) for the continent. We cannot test Tobago's attachment to the

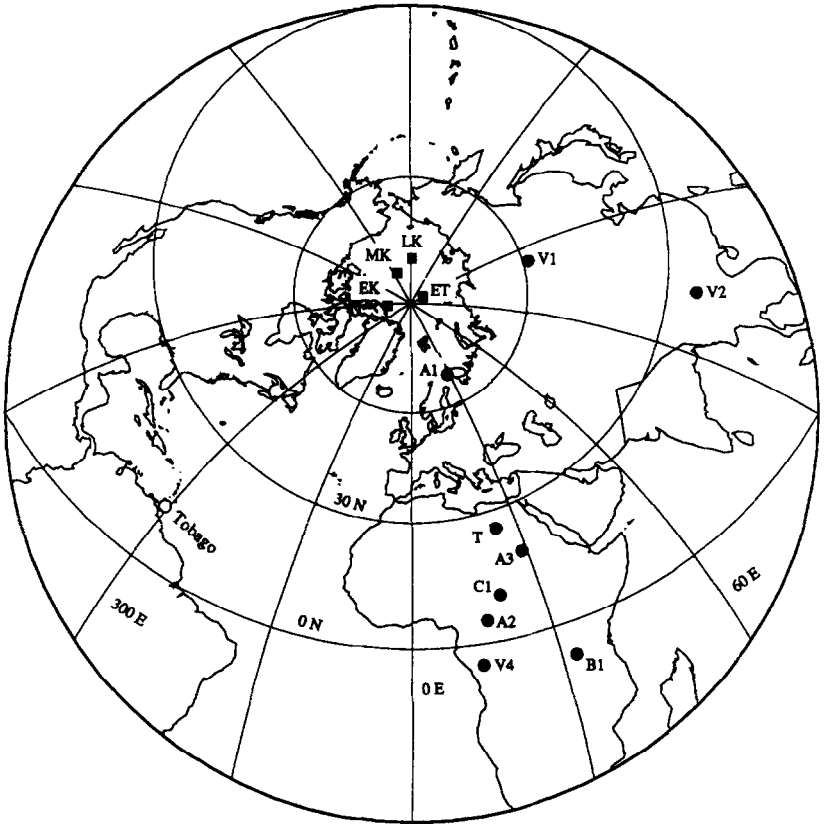


Fig. 6. Equal area projection of paleomagnetic poles from Table 3. The 30°E meridian approximates a small circle around the Caribbean–South American plate boundary zone through the reference and the cluster of observed poles. Poles from the boundary zone arrayed along it (e.g., A1) could have been displaced from a reference pole by vertical axis clockwise rotation.

Caribbean plate because the latter has no reliable APW path. We can, however, compare the Tobago pole and other poles from the plate boundary zone with Cretaceous poles for South America [26–29]. We chose not to use two of the recent poles from South America: [28] because its age is dissimilar to any of the observed poles in Table 3, and [29] because it includes [26] and represents an unspecified time range. To obtain the best age matches we augmented South American poles with poles from North America [30,19] rotated into South American coordinates [20]. We then calculated rotation (R) and poleward motion (P) [31] of the plate boundary zone terranes relative to the reference poles. The poles are plotted in Fig. 6, and tabulated along with R and P in Table 3. Given the large rotations observed, choice of reference pole or age of magnetization are not critical to the interpretation.

Six of the nine poles summarized in Table 3 lie in a well defined cluster that is displaced about 90° from the reference poles. The three outside the cluster are different. Pole A1 comes from the 'predominant' direction measured in the Aruba batholith; Stearns et al. [2] speculated that this may represent a partial chemical remagnetization acquired much later than the nominal age of the intrusion. This interpretation is supported by Fig. 6, which shows that pole A1 lies very near the trajectory defined by rotation of the late Cretaceous South American reference pole about a point located within the plate boundary zone (trajectory approximated by the 30°E meridian). Thus, A1 may simply have acquired its magnetization during a late stage in the rotation process. Poles V1 and V2 diverge strongly. V2 is derived from only two sites, and both V1 and V2 are mainly from intrusive rock and their tectonic corrections may be

in doubt. Some of the six 'cluster' poles are also based on very scanty data, but their close grouping and similarly large rotations (Table 3) is suggestive. Data from Tobago, Aruba and V4 show no significant poleward motion. In contrast, data from B1 and C1 suggest a small amount of northward motion with respect to South America. More work is needed to determine if their poleward displacement has any tectonic significance. Until that is demonstrated, the simplest interpretation of the paleomagnetic data is that Tobago, and probably the other terranes of the central belt, originated near northern South America and have been rotated clockwise subsequently by dextral shear between the Caribbean and South American plates. Perhaps the relatively smaller rotation of Tobago is related to Tobago's eastern position in the plate boundary zone.

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